Problem-Statement:

Design and implement a cloud-native, real-time seismic data processing platform that ingests USGS earthquake events via a streaming pipeline, performs stateful spatiotemporal event correlation to detect potential aftershock clusters, and persists raw, cleansed, and enriched data in a lakehouse architecture. The platform will support real-time cluster detection and historical analytics on seismic magnitude, depth, spatial distribution, and aftershock frequency.

The Idea:

Build a cloud-native system that continuously monitors earthquake events from USGS in real time, automatically groups earthquakes that are close in time and location, and identifies potential aftershock sequences.

How It Works

USGS Earthquake Data

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Real-Time Streaming Pipeline

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Spatiotemporal Correlation

(“Are these earthquakes related?”)

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Potential Aftershock Clusters

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Lakehouse Storage

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Historical Analytics & Pattern Analysis

What the System Provides

- Real-time: Detects and updates earthquake clusters as new events arrive.
- Intelligent: Correlates events based on time, distance, magnitude and depth.
- Analytical: Studies aftershock frequency, spatial distribution and historical patterns.
- Scalable: Cloud-native architecture can handle changes in event volume.

How the 3 categories are used

1. **Real-Time Stream Processing — the engine**

Ingests the USGS feed continuously and runs windowed spatiotemporal correlation on the stream, so clusters are detected and updated the moment new events arrive — not on a scheduled batch job.

2. **Cloud-Native Architecture — the backbone**

Every component (ingestion, correlation engine, dashboard) runs as an independently deployed, containerized microservice on Kubernetes, autoscaling with event volume — so the system stays responsive whether it's a quiet day or an active sequence.

3. **Lakehouse (Lake) Architecture — the memory**

Raw → Cleaned → Enriched layers persist every event and every detected cluster, turning live detections into a queryable historical record for pattern analysis — magnitude, depth, region, decay rate over time.

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Project Tracker Sheet Link :

https://docs.google.com/spreadsheets/d/1t_s9ceWBCumw9T-Ds-D1xpl7wj1Lrhdt4x-Lu2PSHDg/edit?gid=5934887#gid=5934887